STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-HOMOTOPY-COLIMIT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4715-4724 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4726-4742, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 40 of queue_sdga.json (pb=712, stmt 4715-4724, proof 4726-4742).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 090Z: sources/stacks-sdga/pinned-original-source.tex, source lines 4742–4742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 093W: sources/stacks-sdga/pinned-original-source.tex, source lines 4742–4742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FTB: sources/stacks-sdga/pinned-original-source.tex, source lines 4691–4701. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FT9: sources/stacks-sdga/pinned-original-source.tex, source lines 4552–4562. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{M}_n$ be a system of differential graded $\mathcal{A}$-modules. Then the derived colimit hocolim$\mathcal{M}_n$ in $D(\mathcal{A}, d)$ is represented by the differential graded module* $\operatorname{colim} \mathcal{M}_n$.

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Set $\mathcal{M} = \operatorname{colim} \mathcal{M}_n$. We have an exact sequence of differential graded $\mathcal{A}$-modules

$$0 \to \bigoplus \mathcal{M}_n \to \bigoplus \mathcal{M}_n \to \mathcal{M} \to 0$$

by Derived Categories, Lemma 093W (applied the underlying complexes of $\mathcal{O}$-modules). The direct sums are direct sums in $D(\mathcal{A}, \mathrm{d})$ by Lemma 0FT9. Thus the result follows from the definition of derived colimits in Derived Categories, Definition 090Z and the fact that a short exact sequence of complexes gives a distinguished triangle (Lemma 0FTB). □